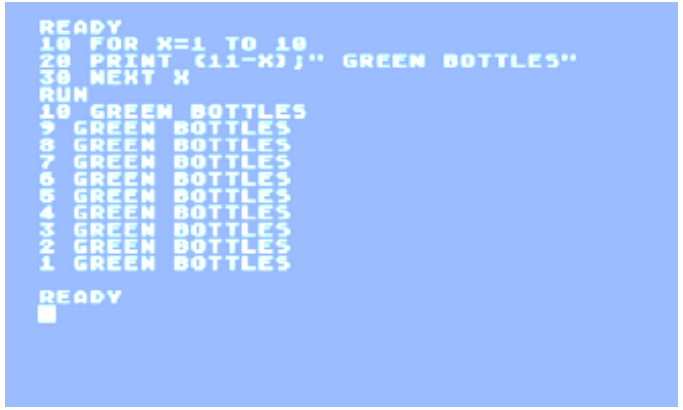
GIS Online Training

SmartNorth – Fall 2026



Coding 101

Rob Hendrickson
GIS Instructor
robbmaps@proton.me

Author: Rob Hendrickson
Data provided by GeoDI Lab
Basemap by MapBox
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Hello!

- My name is Rob Hendrickson (he/him)
  - Master of Geographic Information Science (MGIS)
  - Also a gardener, musician, active community member
  - I do not believe there is such a thing as a “dumb question”
- More on my website, if interested!



Introductions

- Let’s go around the room for quick introductions!
  - Preferred Name & Pronouns
  - What brings you to GIS?
  - Word that begins with same letter as name (eg. Robot Rob)

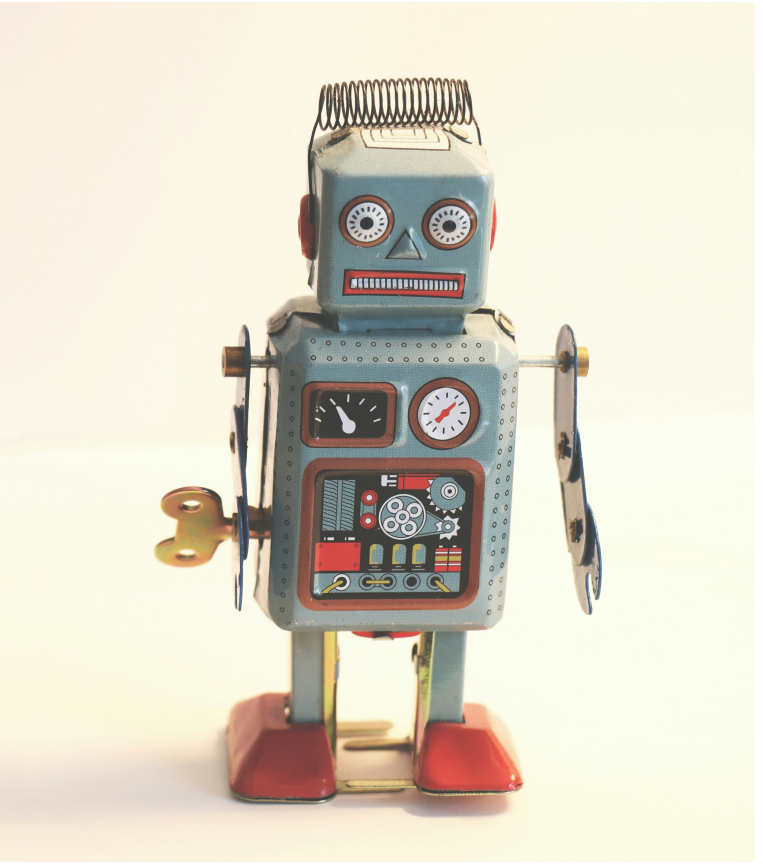


Photo by Emilipothèse on Unsplash

Agenda

What is Coding?

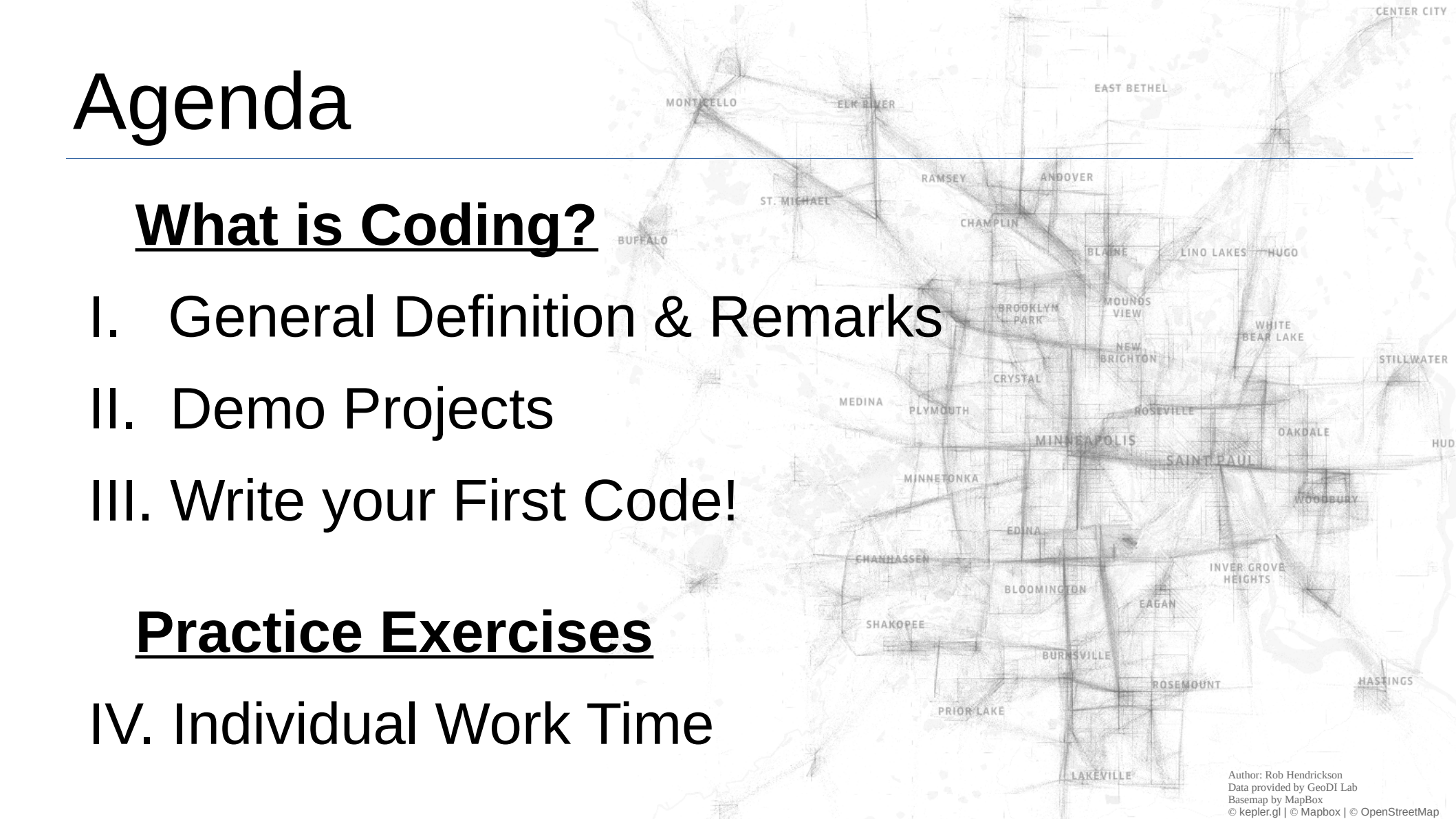
I. General Definition & Remarks

II. Demo Projects

III. Write your First Code!

Practice Exercises

IV. Individual Work Time



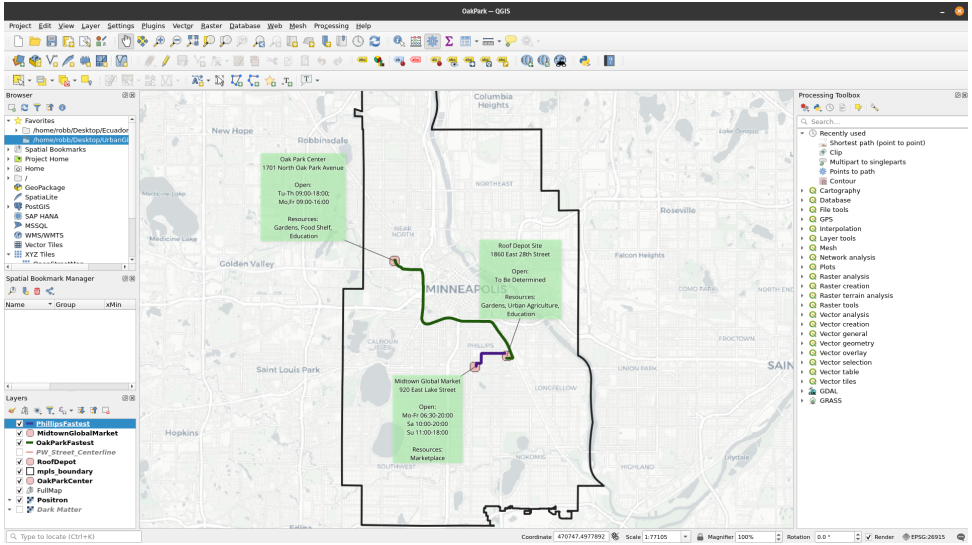
Definition: **GIS**

A Geographic Information System (**GIS**) is a computer-based system to aid in the

<u>collection</u>, <u>maintenance</u>, <u>storage</u>,

<u>analysis</u>, <u>output</u>, and <u>distribution</u>

of spatial data and information





QGIS





From Bolstad (6th Edition)

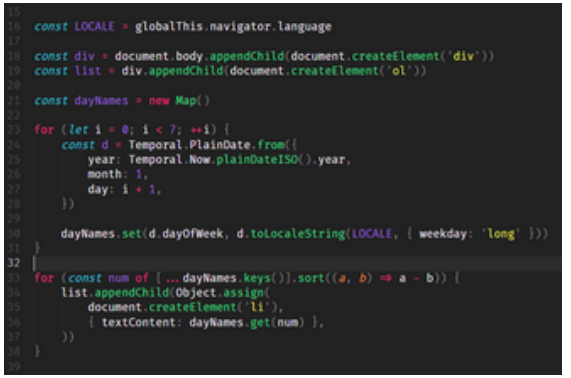
Definition: **Programming**

Programming (AKA coding) is telling a computer what to do!

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- It is composing instructions (programs) that computers can follow



```
const LOCALE = globalThis.navigator.language

const div = document.body.appendChild(document.createElement('div'))
const list = div.appendChild(document.createElement('ol'))

const dayNames = new Map()

for (let i = 0; i < 7; ++i) {
    const d = Temporal.PlainDate.from({
        year: Temporal.Now.plainDateISO().year,
        month: 1,
        day: i + 1,
    })

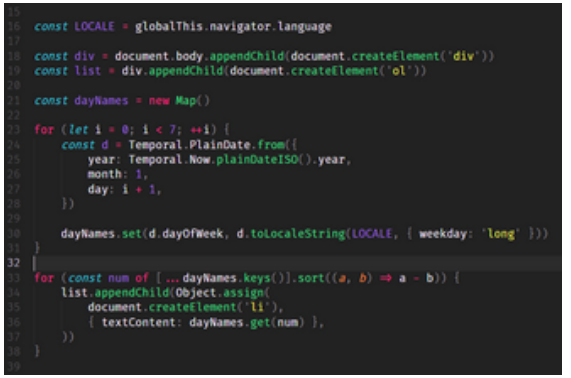
    dayNames.set(d.dayOfWeek, d.toLocaleString(LOCALE, { weekday: 'long' }))
}

for (const num of [...dayNames.keys()].sort((a, b) => a - b)) {
    list.appendChild(Object.assign(
        document.createElement('li'),
        { textContent: dayNames.get(num) },
    ))
}
```

Definition: **Programming**

Programming (AKA coding) is telling a computer what to do!

- It is composing instructions (programs) that computers can follow
- In a programming *language* that computers understand



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JavaScript

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```

JavaScript



Python



R



C's





```
SELECT *
FROM tablename;
```

SQL

```
READY
10 FOR X=1 TO 10
20 PRINT (11-X);" GREEN BOTTLES"
30 NEXT X
RUN
10 GREEN BOTTLES
9 GREEN BOTTLES
8 GREEN BOTTLES
7 GREEN BOTTLES
6 GREEN BOTTLES
5 GREEN BOTTLES
4 GREEN BOTTLES
3 GREEN BOTTLES
2 GREEN BOTTLES
1 GREEN BOTTLES

READY
```

BASIC

Why Coding? - The Three R’s

- Automate a **Repetitive** task
- **Reproduce** an old project from scratch (yours or someone else’s)
- **Reuse** old code for a new project

